Growth applications

1 For each of the following exponential functions, where x represents time:

i Classify as exponential growth or exponential decay.

ii Find the rate of growth or decay per time period as a percentage.

a $y = 0.68\,(1.6)^x$ **b** $y = 0.98\,(0.2)^x$ **c** $y = 100\,(2.05)^x$ **d** $y = 50\,(0.78)^x$

2 The population, P, of a particular town after n years is modelled by $P = P_0\,(1.6)^n$, where P_0 is the original population. Find the population of the town after $3\frac{1}{2}$ years if its original population was $30\,000$. Round your answer to the nearest whole number.

3 The astronomical unit (AU) is often used to measure distances within the solar system. One AU is equal to the average distance between Earth and the Sun, or $92\,955\,630$ miles. The distance, d (in astronomical units), of the nth planet from the Sun can be modeled by the formula:

$$d = \frac{3\left(2^{n-2}\right) + 4}{10}$$

Estimate the distance between Venus and the Sun by using the substitution $n = 2$. Round your answer to two decimal places.

4 The population, A, of aphids in a field of potato plants t weeks after intial observation is modelled by: $A = 1000 \times 2^t$.

a State the initial aphid population.

b Find the predicted aphid population after 5 weeks.

c Find the predicted aphid population 2 weeks before initial observations.

5 Switzerland's population in the next 10 years is expected to grow approximately according to the model $P = 8\,(1 + r)^t$, where P represents the population (in millions) t years from now.
The world population in the next 10 years is expected to grow approximately according to the model $Q = 7130\,(1.0133)^t$, where Q represents the world population (in millions) t years from now.

a Use the model for P to find the current population in Switzerland.

b Use the model for Q to find:

i The current world population.

ii The population of the world in twenty years time, to the nearest million.

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6 The mass in kilograms, M, of a baby orangutan at n months of age is given by the equation $M = 1.8 \times 1.1^n$, for ages up to $n = 6$ months.

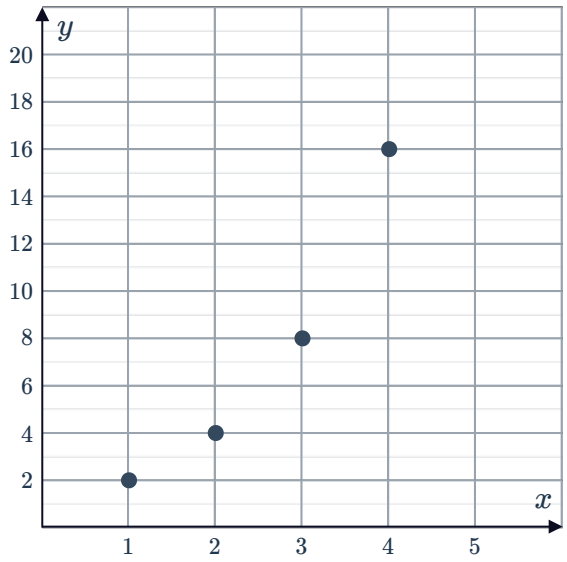
a Find the mass, M, of a baby orangutan at 3 months of age to one decimal place.

b Complete the table, rounding your answers to one decimal place:

Months (n)	0	1	2	3	4	5	6
Mass (M)							

c Sketch a graph of the mass function.

7 Consider the points given in the graph:



a Complete the table of values:

x	1	2	3	4
y				

b Identify the common ratio between consecutive y values.

c State the equation relating x and y.

8 Consider the table of values:

Number of days passed (x)	1	2	3	4	5
Population of shrimp (y)	5	25	125	625	3125

a Is the number of shrimp increasing by the same amount each day?

b Find the equation linking population, y, and time, x, in the form $y = a^x$.

c Sketch a graph of the population over time.

9 Consider the given table of values:

x	1	2	3	4
y	-30	-150	-750	-3750

a Identify the common ratio between consecutive y values.

b Write an equation relating x and y.

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10 A large rectangular piece of paper is folded in half repeatedly. After every fold, the paper is reopened and the number of rectangles counted.

a Complete the table of values for the experiment:

Number of folds (f)	0	1	2	3
Number of rectangles (r)	1			

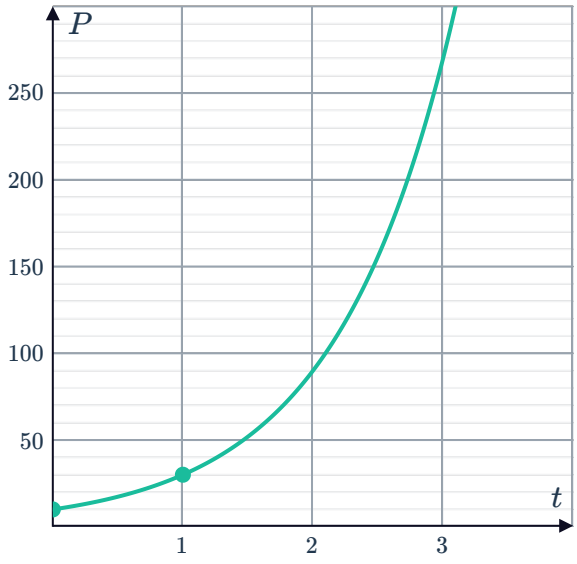
b For each extra fold, the number of rectangles increases by what factor?

c Write an equation for r, the number of rectangles after f folds.

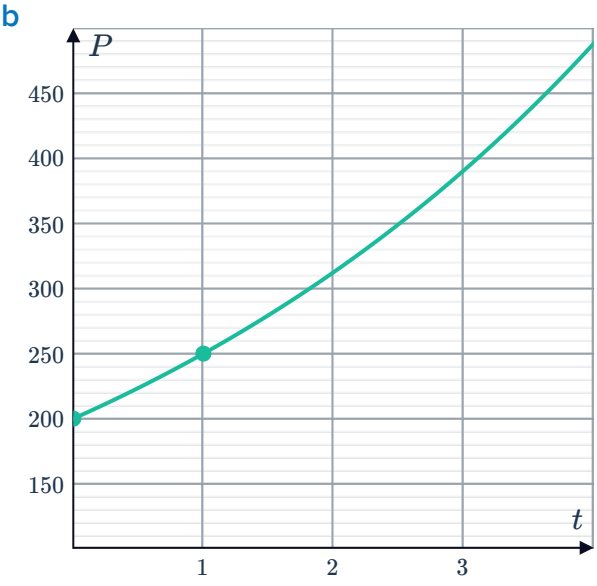
d Find the number of rectangles created by 11 folds.

11 For each of the following graphs of a population, P over time, find the equation of the curve in the form $P(x) = A \times a^x$:

a



b



12 The local rat population numbers approximately 750 and is increasing at a rate of 3% per year.

a Write a function, y, to represent the population after t years.

b Find the estimated size of the population after 5 years.

13 Consider the function $f(t) = \frac{5}{6}(2)^t$, where t represents time.

a Find the initial value of the function as a fraction.

b Express the function in the form $f(t) = \frac{5}{6}(1+r)^t$.

c Does the function represent growth or decay of an amount over time?

d State the rate of growth per time period. Give the rate as a percentage.

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14 The population of a particular mining town increased 160% in 9 years, from 5100 in 2004 to $13\,260$ in 2013. Assume that the population increased at a constant annual rate.

a Find an expression for A, the size of the population y years after 2004. Round your answer to three decimal places.

b Hence, state the annual rate of growth as a percentage, correct to two decimal places.

15 At the start of a particular experiment, there are 2200 bacteria. After x hours, the number of bacteria is given by $A = 2200e^{0.09x}$, where $e \approx 2.7183$.

a A can be represented in the form $A = P(1+r)^x$, where P is the principal amount and r is the hourly rate of growth. Find r in terms of e.

b Rewrite the expression in the form $A = P(1+r)^x$, where r is the hourly rate of growth to four decimal places. Use the approximation $e = 2.7183$.

c State the rate, as a percentage, at which population of bacteria is growing each hour.

16 The number of bacteria in a colony doubles every second.

a Complete the table of values for the first 5 seconds:

Seconds passed (t)	0	1	2	3	4	5
Number of bacteria (y)	1					

b Find an equation for the number of bacteria, y, after t seconds.

c Find the number of bacteria in the colony after 1 minute. Express your answer in exponential form.

d The number of bacteria needs to be graphed against the number of hours h that have passed. Find an equation for the number of bacteria, y, after h hours.

17 The population, P, of a town in millions after t years is approximated by the formula

$$P = 8\left(1 + \frac{2}{100}\right)^t$$

a Complete the table of values, rounding your answers to one decimal place:

t	5	10	15	20	25	30
P						

b Sketch the graph of the function $P = 8\left(1 + \frac{2}{100}\right)^t$.

c After how many years will the population reach 12.6 million?

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18 The number of members at the local club is expected to increase by 19% per year from the current number of 160. The manager of the club is set to receive a bonus when the number of members rises above 727.

a Write a function, y, to represent the number of members after t years.

b Find n, the number of whole years it takes for the number of members at the club to first rise above 727.

19 Kenneth currently works for a company where he sees 90 clients and the number of clients increases by approximately 5 each month. He is considering opening his own business which he knows his 90 clients will follow him to. He anticipates that this will grow by 4% each month.

a Form an equation for P, the number of clients he will have in t months if he stays at the company.

b Form an equation for Q, the number of clients he will have in t months if he opens his own business.

c For which option will he have more clients one year from now?

d Which option will allow him to increase the number of clients at an increasing rate?

20 During a sudden outbreak, scientists must decide between two anti-bacterial treatments that are currently being trialled to try to control the outbreak. In the laboratory, they apply Treatment A and Treatment B to two samples of the bacteria, each containing 300 microbes. They keep track of the number of microbes in each sample. The table shows the results:

Number of hours (t)	0	2	4	6
Number of microbes using Treatment A	300	310	320	330
Number of microbes using Treatment B	300	900	2700	8100

a Which treatment causes the number of microbes to increase linearly?

b By what amount is the number of microbes increasing each hour using Treament A?

c Which treatment will better control the number of microbes?

d Scientists approximate that they'll have a more effective treatment in 10 hours. By that time, what will be the difference between the number of microbes in the two samples?

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21 To accommodate for its expanding population, a country creates a new city and immediately relocates $400\,000$ of its citizens there. The city's land is allocated such that it can immediately produce enough food for $600\,000$ people in the first year. The table shows the functions that can be used to predict the city's population (P) and the number of people who can be fed (Q), after t years:

Population after t years	$P(t) = 400\,000\,(1.013)^t$
Population that can be fed after t years	$Q(t) = 600\,000\left(\frac{t^2}{20\,000} + 1\right)$

a Evaluate $Q(10) - P(10)$ to the nearest whole number.

b Will there be a surplus or shortage of food for the population in 10 years time?

c Evaluate $Q(100) - P(100)$ to the nearest whole number.

d Will there be a surplus or shortage of food for the population in 100 years time?

e Sketch the two functions on the same set of axes over the first 150 years.

f Find the point in time when the models predict the city will have just enough food supplies to support its population. Round your answer to two decimal places.

g After the time found in part (f), will the city have enough food to support its population ever again? Explain your answer.

22 Two mining corporations model the total amount mined, in tonnes, at the end of the nth week of operations. The models are given by:

- Mint Corporation: $M = 100\,(1.15)^{n-1}$
- Crest Corporation: $C = 100n^2$

a Complete the following table of values for the total amount mined by each corporation. Round your answers to the nearest whole number.

n	1	5	10	15
M	100			
C	100			

b Sketch the two functions on the same set of axes over the first 65 weeks of operations.

c If mining operations for both companies were to only last at most a year, which company will have mined the most minerals in that time?

d Find the point(s) of intersection of the two graphs and interpret it in this context.

e At the point of intersection the total quantity of minerals remaining in both mines is equal. If both mining companies continue to operate in the same way indefinitely, which company will exhaust their mine first?

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Decay applications

23 The local rodent population, numbering approximately 840, is decreasing at a rate of 4% per year.

a Write a function, y, to represent the population after m years.

b Find the size of the population after 8 years.

24 The number of taekwondo clubs worldwide was approximately 2310 in 1988 and has been decreasing at a rate of 2.6% per year since.

a Write a function, y, to represent the number of taekwondo clubs n years after 1988.

b Find the number of taekwondo clubs in 2009.

25 A ball dropped from a height of $21\ \text{m}$ will bounce back off the ground to 50% of the height of the previous bounce (or the height from which it is dropped when considering the first bounce).

a Write a function, y, to represent the height of the nth bounce.

b Calculate the height of the fifth bounce, correct to two decimal places.

26 Two people are asked to eat the same food and in equal quantity. They were chosen such that Person A incorporates lots of sugar into their diet while person B consumes limited amounts of sugar in their diet.
After eating, their blood sugar level peaked, and was then continually measured, with the following functions modelling their blood sugar level at time t hours:

- Person A: $f(t) = 179(0.3)^t$
- Person B: $g(t) = 132(0.6)^t$

a State the highest blood sugar level measured for:

i Person A. **ii** Person B.

b Find the rate, as a percentage, that the blood sugar level decreased each hour for:

i Person A. **ii** Person B.

c Comment on how sugar affects the blood of people who have a high sugar diet compared to a low sugar diet.

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27 When a heated substance such as water starts to cool, its temperature T at time t minutes after being left to cool is given by an exponential function.

a Should T be an increasing or decreasing function?

b For positive values of a, state whether the following functions could be used to model the temperature T at time t:

i $T = -ab^t$, where $b > 1$.

ii $T = ab^{-t}$, where $b > 1$.

iii $T = ab^t$, where $0 \leq b \leq 1$.

iv $T = -ab^t$, where $0 \leq b \leq 1$.

c What does b represent in the model?

d Substance P starts off at a temperature of 145°C and is left to cool in a room whose temperature is a constant 0°C. By using the table below, find the equation for the temperature of the substance.

Minutes passed (t)	0	1	2	3	4
Temperature (T)	145	87	52.2	31.32	18.792

e Find the temperature after 10 minutes, correct to two decimal places.

f Will the temperature of the substance ever reach 0°C?

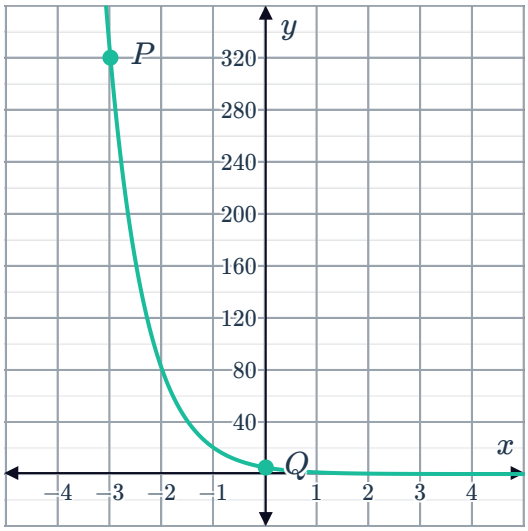
g The temperature of another substance Q which also starts off at a temperature of 145°C is modelled by $T = 145 \times 0.9^t$. Which substance will cool more rapidly?

28 The points $P\,(-3, 320)$ and $Q\,(0, 5)$ satisfy an exponential function of the form $y = ab^{-x}$.

a Find the value of a.

b Find the value of b.

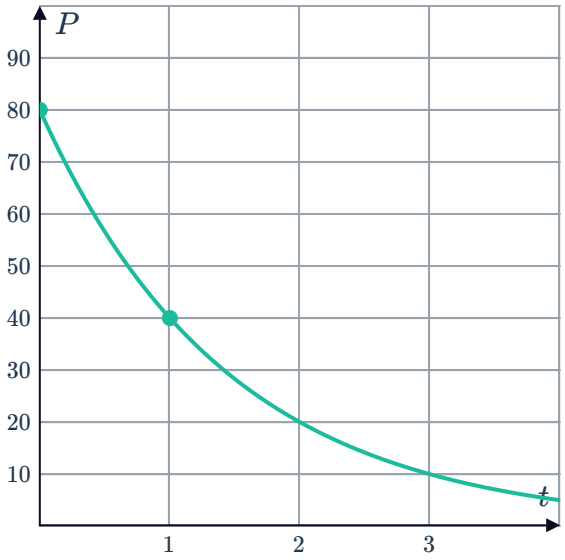
c Hence, find the value of y when $x = 2$.



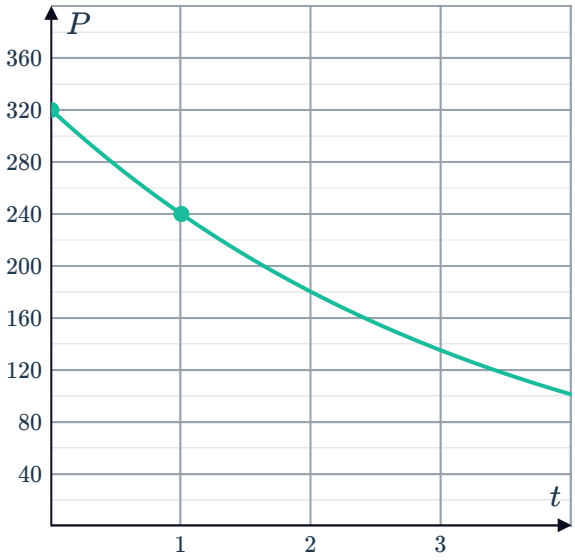
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29 For each of the following graphs of a population, P over time, find the equation of the curve in the form $P(x) = A \times a^x$:

a



b



30 Consider the function $f(t) = 2\left(\frac{3}{8}\right)^t$, where t represents time.

a Find the initial value of the function.

b Express the function in the form $f(t) = 2(1-r)^t$, where r is a decimal.

c Does the function represent growth or decay of an amount over time?

d State the rate of change per time period as a percentage.

31 A sample contains 60 grams of iodine-131, which has a half-life of 8 days.

a Write a function, A, to represent the amount of iodine-131 remaining in the sample after n days.

b Find the amount of the isotope left after 72 days. Round your answer to two decimal places.

32 A sample contains 300 grams of carbon-11, which has a half-life of 20 minutes.

a Write a function, A, to represent the amount of carbon-11 remaining in the sample after n minutes.

b Find the amount of the isotope left after after 3 hours. Round your answer to two decimal places.

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33 Carbon-14 is an element with a half-life of 5730 years that is found in all living plants and animals. When the plant or animal dies, carbon-14 is no longer taken in, and as such, it is useful in estimating the age of fossils. Suppose the fossil of an organism originally with 385 grams of carbon-14 has been found.

a Write a function, A, to represent the amount of carbon-14 remaining in the fossil after n years.

b Find how much carbon-14 would be left in the fossil today if the organism died 1500 years ago. Round your answer to two decimal places.

c Find how long ago the organism died if the fossil has 35 grams of carbon-14 remaining. Round your answer to the nearest number of years.

34 Starting at k grams, the amount of chromium-51 in a sample after $\frac{t}{28}$ days is given by:

$$A = k\left(\frac{1}{2}\right)^{\frac{t}{28}}$$

a Rewrite the model in the form $A = P(1-r)^t$, where r is the daily decay rate correct to four decimal places.

b State the daily decay rate as a percentage, correct to two decimal places.

35 At the start of an experiment, there are 8900 grams of a particular radioactive isotope. The number of grams left of the isotope after t years is given by $A = 8900e^{-0.08t}$, where $e \approx 2.7183$.

a Rewrite the expression in the form $A = P(1-r)^t$, where r is the annual rate of decay to four decimal places. Use the approximation $e = 2.7183$.

b State the annual decay rate as a percentage, correct to two decimal places.

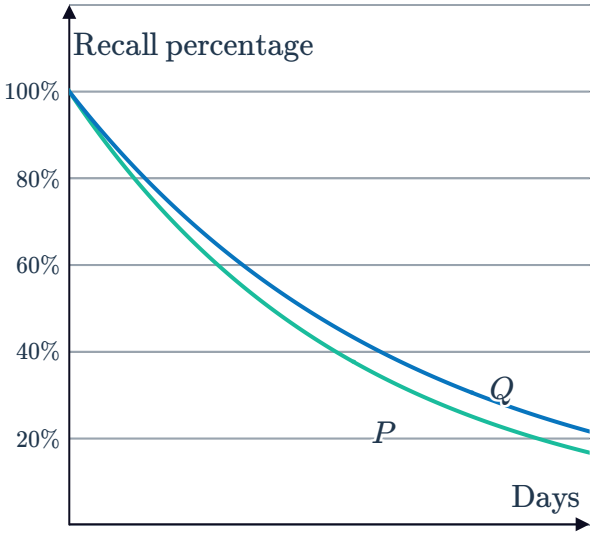
36 The population of a particular mining town decreased 80% in 7 years from 3400 in 2004 to 680 in 2011. Assume a constant annual rate of decrease.

a Find the expression for A, the size of the population t years after 2004. Give your answer in the form $A = P(1-r)^t$, where r is the annual rate of decrease correct to four decimal places.

b State the annual rate of decrease as a percentage, correct to two decimal places.

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37 In a memory study, subjects are asked to memorise some content and recall as much as they can remember each day thereafter. The given graphs represent the percentage of content remembered by two participants Maximilian (P) and Lucy (Q) after t days:



a Each day, Maximilian finds that he has forgotten 15% of what he could recount the day before. Form a model for the percentage P that Maximilian can still recall after t days.

b Who was forgetting more rapidly?

c Given one of the following is the model for the percentage that Lucy can still recall after t days, which model is it?

A $Q = (0.85)^t$ **B** $Q = (0.83)^t$

C $Q = (0.87)^t$ **D** $Q = (0.81)^t$

d On each day, what percentage of the previous day's content did Lucy forget?

e According to the models, will either of them completely forget the content?

38 A farmer currently runs operations over a 200 acre portion of his land. He is looking to slowly wind up operations on his farm so that he can rehabilitate the land from overuse. Rather than stopping operations all at once, he decides to extend the existing farming area by 2% over the course of each year and fence off 5.7% of the total area at the end of each year for rehabilitation.

a Find the area of land he will have to farm on at the beginning of the second year.

b Find the area of land he will have to farm on at the beginning of the third year. Round your answer to two decimal places.

c Find the function $f(n)$ which models the area of land available for farming at the beginning of the nth year.

d Find the overall percentage rate the area of farming land is decreasing by each year.

Financial applications

39 Justin purchased a piece of sports memorabilia for $\$2900$, and it is expected to appreciate in value by 9% per year.

a Write a function, y, to represent the value of the piece of sports memorabilia after t years.

b Find the estimated value of the piece of sports memorabilia after 8 years.

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40 Find the value of $P\left(1+\frac{r}{k}\right)^{kn}$ for $P=3000$, $r=5\%$, $k=4$ and $n=2$ to two decimal places.

41 A fixed-rate investment generates a return of 6% per annum, compounded annually. The value of the investment is modelled by $A=P\,(1.06)^t$, where P is the original investment. Find the value of the investment after after $3\frac{1}{4}$ years if the original investment was $\$200$.

42 The annual interest rate for a savings account is 10%. Write an expression that gives the value of $\$1$ after t years, if the rate is compounded:

a Annually.

b Monthly.

43 For each of the following investments:

i Write a function, A, to model the value of his investment after t years.

ii Find the value of the investment after 5 years.

a $\$6100$ invested at 8.3% p.a. compounded annually.

b $\$6800$ invested at 3% p.a. compounded monthly.

c $\$3200$ invested at 5.2% p.a. compounded quarterly.

44 The equation $A=3000\times 1.12^n$ can be used to calculate the value A, of a $\$3000$ investment at 12% p.a. with interest compounded annually, after n years.

a Sketch the graph of this function for $0\leq t\leq 10$.

b What would the equation be if interest was compounded semiannually over n years?

c What would the equation be if interest was compounded monthly over n years?

d Add the graphs of the functions from parts (b) and (c) to your graph from part (a).

e Find the difference in the value of the investment at the end of 10 years if the interest was compounded monthly compared to annually.

45 Vanessa invests $\$1000$ into a term deposit, which is compounded at a rate r each year, as shown in the following table:

Years passed (n)	0	1	2
Value of investment $(\$A)$	1000	1020	1040.40

a Find the value of r from the table of values.

b Write a function, A, to model the value of the investment after n years.

c Find the value of the investment after 6 years, assuming the rate stays constant.

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46 Mohamad has just started a new job, where his starting salary is $\$50\,000$ p.a. and is expected to increase by 3.2% each year. Elizabeth has also just started a new job, which has a starting salary of $\$49\,000$ p.a. and is expected to increase at a rate of 6.1% each year.

a Write an equation for M, Mohamad's salary in t years time.

b Write an equation for P, Elizabeth's salary in t years time.

c How much greater will Elizabeth's salary be in 6 years' time than Mohamad's?

d Over the long term, whose salary will increase more rapidly?

47 If the electricity bill is not paid by the due date, the company charges a fee for each day that it is overdue. The table shows the fees:

Number of days after bill due (x)	1	2	3	4	5	6
Overdue fee in dollars (y)	4	8	16	32	64	128

a At what rate is the fee increasing each day?

b State the overdue fee the bill will incur if it is paid 5 days late.

c Write a function that models the overdue fee y as a function of the number of days overdue x.

48 When comparing two phone companies, Derek notices that both offer a $\$200$ monthly contract, but they have different policies on how they charge fees if the bill is overdue:

- Company A: Start adding $\$6$ to the bill for every day the bill is overdue.
- Company B: Increase the bill by 4% per day that the bill is overdue.

a Complete the table of values for the total amount of the bill if it is x days overdue under each company's policy.

x	0	3	6	9
Company A's bill	200			
Company B's bill	200			

b If an unpaid bill reaches $\$3000$, it is referred to a debt collector. Under which company's policy would an unpaid bill first reach $\$3000$?

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49 Sophia is considering two different career options:

- She can accept a job with a company that has offered her a salary of $\$125\,000$ a year with a projected salary increase of $\$3000$ per year.
- She can start her own business where she estimates her initial annual salary to be $\$20\,000$ per year with a projected annual salary increase of 40% each year.

a Form an equation for $f(n)$, her salary in n years time if she accepts the job with a company.

b Form an equation for $g(n)$, her salary in n years time if she starts her own business.

c Complete the table of values:

n	2	3	4
Increase in salary compared to previous year when working for a company			
Increase in salary compared to previous year when working for her own business			

d After 3 years, which option will result in a higher annual salary for Sophia?

e After 6 years, which option will result in a higher annual salary for Sophia?

50 Bank A pays interest at a rate of 5.3% p.a. compounded annually, while Bank B offers a rate of 5% p.a. with interest compounded quarterly.

a Form an equation for A, the amount $\$P$ would grow to after t years if invested with:

i Bank A. **ii** Bank B.

b State the effective annual rate for Bank B as a percentage, to two decimal places.

c Which bank offers the better deal?

51 A savings account with a particular bank pays interest compounded annually at a rate of 19% per annum. Therefore, the amount that $\$17\,000$ has grown to n years after it is deposited into a savings account is given by: $A = 17\,000\,(1.19)^n$.

a Create a model in terms of n to represent the value of the investment after a certain number of quarters by rewriting the equation in the form $A = P\,(1+r)^{kn}$, where r is the effective quarterly interest rate correct to four decimal places and k is a constant.

b State the effective quarterly interest rate as a percentage, correct to two decimal places.

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52 Iain is trying to decide between two investments to put his $\$600$ savings in:

- An online savings account that offers 0.46% per month interest compounded monthly.
- A corporate bond that pays 4.80% per annum interest compounded annually.

a Form an equation for A, the amount $\$600$ would grow to after t years if invested with:

i The online savings account. **ii** The corporate bond.

b Rewrite the equation for the corporate bond in the form $A = P(1+r)^{kn}$, where r is the effective monthly rate correct to four decimal places and k is a constant.

c State the effective monthly interest rate for the investment in the corporate bond as a percentage, correct to two decimal places.

d Which option has the higher return?

53 Due to favourable market conditions, a company's annual costs are expected to decrease by 5% per year for the next few years. Their annual costs for the current financial year are $\$322\,000$.

a Write a function, y, to represent the company's annual costs in m years.

b Find the company's annual costs in 5 years.

54 The median national sale price of property over the next 10 years can be approximated by the equation $P = 621\,900\,(0.907)^t$, where P represents the median sale price t years from now.
The median national rental price of property over the next 10 years can be approximated by the equation $Q = 320\,(1.053)^t$, where Q represents the median rental price t years from now.

a According to the models, which will decrease over the next 10 years?

b At what annual percentage rate are rental prices expected to increase over the next 10 years?

c If interest rates decrease at any time, the annual percentage change in sale prices is expected to increase by 4%. If this occurs, what will be the new annual percentage change in sale prices?

55 A property has an original value of P and doubles in value every 14 years. Find the annual rate of growth, as a percentage to two decimal places.

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56 An economist expects that over the next 12 years, the value of the US dollar (relative to the Australian dollar) will change according to the formula:

$$f(t) = \frac{127\,(92^t)}{10^{2(t+1)}}$$

Where t represents the number of years from now.

a Express the formula in the form $f(t) = A(B)^t$, where A and B are decimals.

b By what percentage is the value of the US dollar expected to decrease each year?

c Find the value of the US dollar, relative to the Australian dollar in 12 years time.

57 A car originally valued at $\$28\,000$ is depreciated at the rate of 15% per year. The salvage value, S, of the car after n years is given by: $S = 28\,000\,(0.85)^n$.

a Complete the following table, rounding your answers to two decimal places:

n	2	4	6	8	10	12
S						

b Sketch the graph of the salvage value of the car over 12 years.

c Find the value of the car after 3 years.

d After how many years does the value of the car drop down to an amount of $\$6\,485.27$?

58 A van originally valued at $\$45\,000$ depreciates at a rate of 22% per year.

a Write the rule for A, the value of the van t years after it is purchased.

b Find the value of the van after 4 years.

59 A new car purchased for $\$38\,200$ depreciates at a rate $r\%$ each year. The value of the car for the first two years is shown in the table below:

Years passed (n)	0	1	2
Value of car $(\$A)$	$38\,200$	$37\,818$	$37\,439.82$

a Find the value of r.

b Write the rule for A, the value of the car n years after it is purchased.

c Find the values of the car after 6 years.

d A new motorbike purchased for the same amount depreciates according to the model $V = 38\,200 \times 0.97^n$. Which vehicle depreciates more rapidly?

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60 The value of a $\$25\,000$ motorbike t years after it is purchased is given by $A = 25\,000\,(0.85)^t$.

a Create a model in terms of t to represent the value of the motorbike after a certain number of weeks by rewriting the equation in the form $A = P\,(1 - r)^{kt}$, where r is the effective weekly depreciation rate correct to four decimal places and k is a constant.

b State the effective weekly depreciation rate as a percentage, to two decimal places.

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